Chapter 18

Applications Of Inversive Geometry

New Proofs Of Existing Theorems

Generating sufficiently distinct proofs of existing theorems is a fun and useful exercise that has its own academic merit for the following two reasons. First, when we embark on a journey of generating a new proof of a mathematical statement we face an open-ended problem, which is the type of problems that practicing mathematicians deal with on the regular bases.

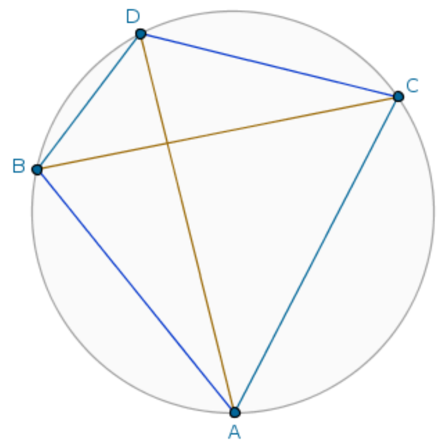
Second, when, in our mind, we enumerate and zip through the potential candidate layers whose machinery may help us generate such a new proof we precisely *learn and study mathematics*. And even if in the end we fail to construct any new proof at all we still gain in the study of mathematics because a failure in mathematics is just as, if not more, educational than a success.

As such, now that we have learned enough of the relevant theoretical material, let us see if we can generate a sufficiently distinct proof of the Ptolemy's Theorem that bravely plows its path exactly through the machinery of inversive geometry.

According to that theorem:

in an arbitrary cyclic quadrilateral the sum of the products of the lengths of the opposite sides is equal to the product of the lengths of the diagonals

Pictorially, if $ACDB$ is a cyclic quadrilateral (Figure 18.1):



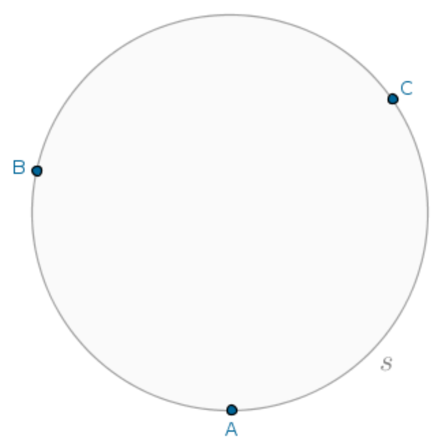
then it is the case that:

$$|AD| \cdot |CB| = |AC| \cdot |DB| + |AB| \cdot |CD| \tag{1}$$

and our problem officially becomes

Problem 24. Prove the Ptolemy's Theorem using inversion of the real plane in a circle.

Proof. It can be shown separately that through any *three* distinct non-collinear points (Figure 18.2):

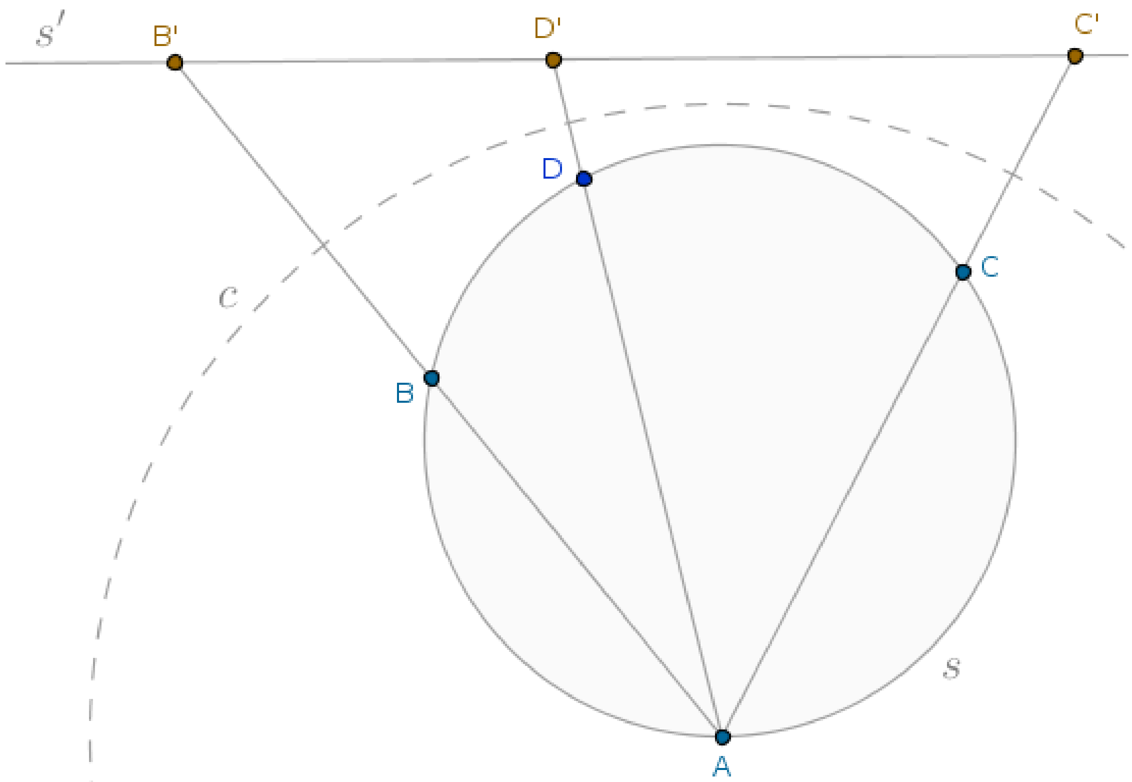


such as the points A, B and C, there passes a unique circle, such as the circle s in the Figure 18.2 above.

That fact alone is already very helpful but the mere addition of a single point, name that point D, throws a monkey wrench into the delicate works that our circle s is because that point is under no obligation to sit on the same circle.

Thus, let us consider the inversion of the parent plane that houses all *four* points of interest with respect to a circle, $c(A, r)$, centered at the point, say, A with the radius r that is strictly larger than the diameter, d_s, of the circle s: $r > d_s$.

Then, due to the Theorem 19, we know that under the said inversion, with positive power, the unique circle s that carries our quadrilateral $ACDB$ will be taken into a straight line, s', such that that straight line will have exactly zero points in common with the circle s because we agreed that $r > d_s$ (Figure 18.3):



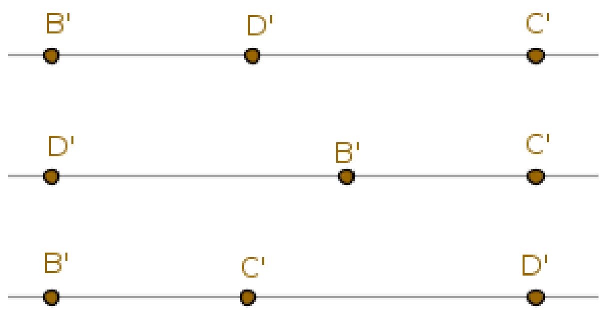
Moreover, the points B and C will be taken into their inverse images B' and C', each located on the straight line s'.

Now we take the fourth point, D, in the same plane and distinct from the points A, B and C. Let D' be the inverse image of the point D under the inversion of the parent plane with respect to the circle $c(A, r)$

. If the point D is located on the circle s then the point D' will be located on the straight line s'. Inversely, if the point D is **not** located on the circle s then the point D' will **not** be located on the straight line s'.

Hence, the four points A, B, C and D belong to the same circle s if and only if the three image points B', C' and D' belong to the same straight line, s', under the transformation described.

Now let us agree that while the points B and C are fixed, the location of the point D with respect to the points B and C can vary in a very small number of possible permutations (Figure 18.4):



The above arrangement of points is a typical set up of the notion of *in-betweenness* in the Hilbert's axioms of geometry. As such, for the first arrangement of points we have:

$$|B'D'| + |D'C'| = |B'C'| \tag{2}$$

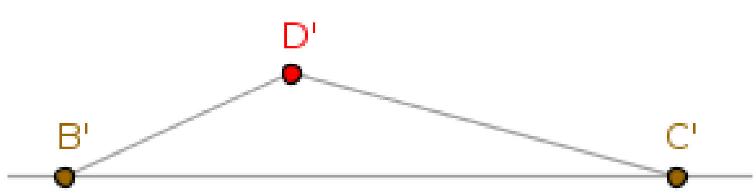
For the second arrangement of points we have:

$$|D'B'| + |B'C'| = |D'C'| \tag{3}$$

For the third arrangement of points we have:

$$|B'C'| + |C'D'| = |B'D'| \tag{4}$$

Observe that the above equations are the embodiment of the notion of *a minimizing geodesic* or a curve that minimizes the distance between any two distinct points in the given space (Figure 18.5):



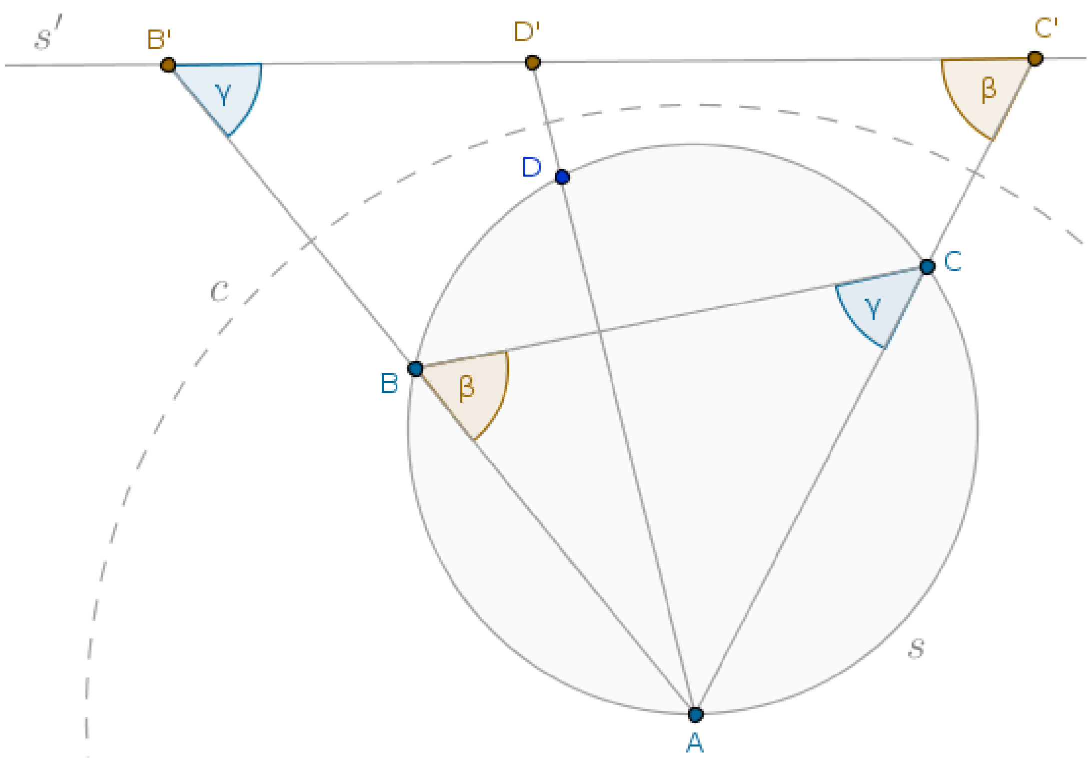
The shortest path between any two points in a Euclidean plane is a straight line. Thus, if the point D' is located not on the straight line s' then, according to the triangle inequality, all of the above equalities will fail to hold in which case for the relevant distances we will have the following relationship:

$$|B'D'| + |D'C'| > |B'C'| \tag{5}$$

Moreover, according to the “Transform And Conquer” problem-solving approach, the distance relations shown above represent a solution of the transformed problem, or a solution that is supposed to be *easier* in some sense than the original. Now let us investigate what these transformed expressions will become if, as the theory goes, we reverse-transform them into the original setting.

That is, let us rewrite the expression in, say, (**2**) in such a way that only the original points A, B, C and D figure into it.

Applying the Theorem 22 (ICSTT) to the triangles ABC and $AC'B'$ (Figure 18.6):



we see that these triangles are similar.

Here, we would like to direct the reader’s attention to the fact that our designation of similar triangles is not haphazard but is very meaningful and important, as we arrange the names of the vertices in such triangles in such a way that the magnitudes of the interior angles of the corresponding triangles at the corresponding vertices are equal one another.

In this particular case:

$$\angle BAC = \angle C'AB'$$

$$\angle ABC = \beta = \angle AC'B'$$

$$\angle ACB = \gamma = \angle AB'C'$$

Then, the order of the names of the vertices in such similar triangles tells us right away which interior angles are equal across the two triangles and the lengths of which sides across these triangles are in the same proportion.

From now on, we will assume that the readers understand that the order of the names of the vertices of similar triangles can not be ignored. As such, from the fact that the triangles ABC and $AC'B'$ are similar, it follows that:

$$\frac{|B'C'|}{|AB'|} = \frac{|CB|}{|AC|}$$

from where:

$$|B'C'| = \frac{|CB| \cdot |AB'|}{|AC|}$$

But from the Condition 3 of Definition 1 we have it that:

$$|AB| \cdot |AB'| = r^2$$

Hence:

$$|B'C'| = \frac{|CB| \cdot r^2}{|AB| \cdot |AC|} \tag{6}$$

Likewise, from the similarity of the remaining two pairs of triangles $ABD/AD'B'$ and $ADC/AC'D'$, it follows that:

$$|B'D'| = \frac{|DB| \cdot r^2}{|AB| \cdot |AD|} \tag{7}$$

and, correspondingly:

$$|D'C'| = \frac{|CD| \cdot r^2}{|AD| \cdot |AC|} \tag{8}$$

Putting (**6**, **7**, **8**) back into (**2**), we find:

$$\frac{|CB| \cdot r^2}{|AB| \cdot |AC|} = \frac{|DB| \cdot r^2}{|AB| \cdot |AD|} + \frac{|CD| \cdot r^2}{|AD| \cdot |AC|}$$

Since it is the case that $r > 0$, it follows that:

$$\frac{|CB|}{|AB| \cdot |AC|} = \frac{|DB|}{|AB| \cdot |AD|} + \frac{|CD|}{|AD| \cdot |AC|}$$

Multiplying the above relation through by $|AB| \cdot |AD| \cdot |AC|$, we arrive at:

$$|DB| \cdot |AC| + |CD| \cdot |AB| = |CB| \cdot |AD| \tag{9}$$

On the other hand, if the point D' is located not on the straight line s' then it will be the case that:

$$|DB| \cdot |AC| + |CD| \cdot |AB| > |CB| \cdot |AD| \tag{10}$$

Thus, what remains to be done is to just state more formally the following

Theorem. If the four distinct points A, B, C and D are located on the same circle in such a way that the points B and C belong to different arcs with the end points A and D then it is the case that:

$$|DB| \cdot |AC| + |CD| \cdot |AB| = |CB| \cdot |AD|$$

while if these points A, B, C and D are located not on the same circle then it is the case that:

$$|DB| \cdot |AC| + |CD| \cdot |AB| > |CB| \cdot |AD|$$

We, thus, proved that IF P THEN Q and that IF NOT P THEN NOT Q. But the last implication is logically equivalent to IF NOT NOT Q THEN NOT NOT P. Which is to say that the last implication is logically equivalent to IF Q THEN P. Thus, we proved both IF P THEN Q and IF Q THEN P.

Hence, we actually proved the following

Theorem. The four distinct points A, B, C and D are located on the same circle in such a way that the points B and C belong to different arcs with the endpoints A and D if and only if the following relation:

$$|AC| \cdot |DB| + |CD| \cdot |AB| = |CB| \cdot |AD| \tag{11}$$

holds.

But if the points A, B, C and D are located on the same circle then the corresponding quadrilateral $ACDB$, whose vertices are specified *in that order*, is cyclic and it is the case that:

$$|AD| \cdot |CB| = |AC| \cdot |DB| + |AB| \cdot |CD|$$

as was to be shown. □

A short analysis of what just transpired: notice again how the transformed solution of the given problem captured in (**2**) was next to trivial or next to childish to construct or obtain. Such a perceived tactical ease was afforded by the application of a successful transformation which is the point of the main idea that we have outlined in Chapter 10.

Thus, one more time we witnessed the intuitive:

transform-solve-untransform

heuristic template in action.

Once a successful transformation was found, we have spent most of our time babysitting the sides of similar triangles that are in the same proportion.